

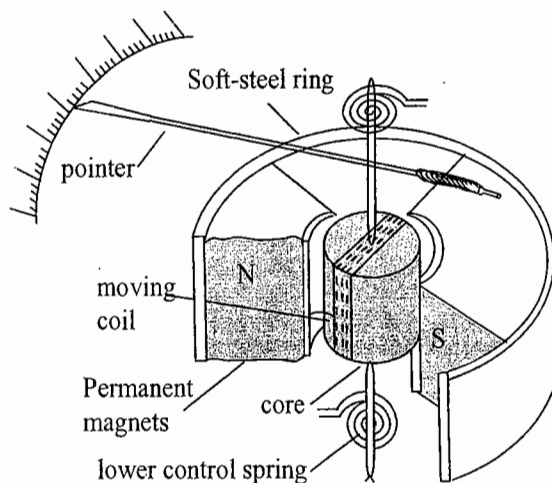
Electromechanical Indicating Instruments

Analog ammeter, voltmeter and ohm meter

Types of instruments used for measurements of current and voltage

- PMMC (Permanent magnet moving coil)
- Moving iron type (AC & DC)
- Electrodynamometer type (AC & DC)
- Electro thermic instrument (AC & DC)
Eg: (Hot wire, thermocouple)
- Electrostatic type (AC & DC)

PMMC: (Permanent magnet moving coil/ D'Arsonal instrument:



- It is similar to a permanent magnet DC motor with a limited rotation
- It works whenever a current carrying conductor is placed in a magnetic field. Experiences a mechanical force according to Lorentz's principle
 $T_d = BINA \text{ N-m}$ ($\theta = 90^\circ$ for max)
- Springs will carry the current

- Spring control is used to produce controlling torque springs are made up of bronze
Note: Radial magnetic field concept is used
- Eddy current damping is used (Al frame produce damping torque) for voltmeter
- In ammeter moving coil is wound on aluminium former, aluminium former produces damping torque.

Torque produced in PMMC:

$$T_d = BINA$$

$$T_c = k_c \theta$$

At steady state position ($T_c = T_d$) not ($T_d = T_c$)

$$T_c = T_d$$

$$k_c \theta = BINA$$

$$\theta = \frac{BINA}{k_c} \quad (\because B, N, A, k_c = \text{constant})$$

$\theta \propto I \rightarrow$ Hence scale in PMMC is linear.

- Hence with ageing effect of spring the reading will be more

$$\theta \propto A$$

$$\theta \propto B$$

$$\theta \propto N$$

Hence with the decreasing field of magnets, reading will be less.

Note: M.C instrument works only for DC

Example 1:

A 1 - ϕ , 230V 50Hz supply is given to a (0-300V) m.c instrument then it reads '0 V'.

Sol: If high frequency ac supply is given, for positive half cycle pointer moves zero to final value and final value to zero. And for negative half cycle current drawn is reversed and then pointer moves below zero. And



shows the '0' reading because of stoppers like this is to be happen '50' times in one second. So because of inertia the pointer stays '0' position only and vibrates there.

Note: If low frequency ac supply (say 1 Hz) then the pointer oscillates between '0' and final reading continuously

- When mc instrument connected in series with the load then deflection (θ) is proportional to the current passing through the load. Hence it works as a ammeter
- When mc instrument connected across the load then the current passing through meter proportional to potential difference across two terminals of meter

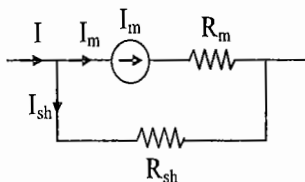
$$\theta \propto V$$

Moving coil instruments average values of current and voltage

Extension range of PMMC:

- Measuring the higher values of dc currents voltages by a small circuit modifications in the existing instruments is called extension of range of an instrument

1. dc ammeter:



$$I_{sh} = (I - I_m)$$

$$I_m R_m = I_{sh} R_{sh}$$

$$R_{sh} = \frac{R_m}{m-1} \Rightarrow m = \frac{I}{I_m}$$

'm' is called multiplication factor

- To increase range of an ammeter, A low value of Resistance must be connected in parallel with the meter. It is called as shunt resistance (R_{sh})
- The internal resistance (R_m) of an ammeter ideally zero ohm (sc).

Note: The ammeter must be connected in series with the load. If it is connected in parallel with the load the supply voltage appear across the ammeter and also because of its low internal resistance meter draws more current and meter will be damaged

Example 2:

(0 – 1 mA) ammeter has its interval resistance of 4Ω . It's range is to be extended to 100 A. Then the value of shunt resistance required is

$$\text{Sol: } I = 100\text{A} \quad m = \frac{I}{I_m} = 100 \text{ k}$$

$$\begin{aligned} I_m &= 1 \text{ mA} & R_{sh} &= \frac{R_m}{m-1} \\ & & &= \frac{4}{100\text{k}-1} \\ & & &= 40 \mu\Omega \end{aligned}$$

Example 3:

If an analog PMMC (0–10) A ammeter is provided with no controlling mechanism and the moving parts are free to rotate, the reading of the instrument if a current of 1A DC Passed through the moving coil.

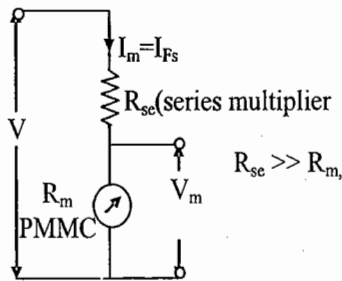
- (a) The pointer will remain stationary
(b) The pointer will continuously rotate
(c) 1A
(d) 10 A

Ans: (b)



Sol: Since there is no control mechanism, T_C will not develop and T_d will develop because coil is carrying current.

Voltmeter Series Multipliers



$$V_m = \frac{V R_m}{R_{sc} + R_m}$$

$$m = \frac{V}{V_m} = \text{multiplying factor}$$

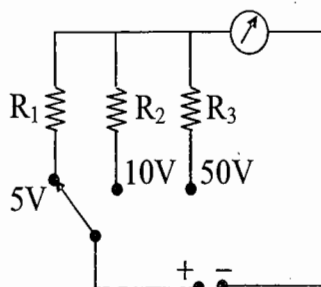
$$m = \frac{V}{V_m} = \frac{\text{Fullscale of voltage}}{\text{measure value of voltage}}$$

$$\text{Voltmeter sensitivity} = S_v = \frac{1}{I_{fs}} = \frac{R_{sc} + R_m}{V} \Omega$$

- Our human body resistance is 600 - 1000Ω

Example 4:

A multi-range voltmeter is shown in the figure with full scale deflection current of $50\mu A$ and meter resistance of $1K\Omega$. What are the values of resistances R_1 , R_2 & R_3 respectively?



- (a) 100 KΩ, 200 KΩ, 1000 kΩ
- (b) 101 KΩ, 201 KΩ, 1001 kΩ
- (c) 99 KΩ, 199 KΩ, 999 kΩ
- (d) 5 KΩ, 10 KΩ, 50 kΩ

Ans: (c)

$$\text{Sol: } R_1 = R_m \left[\frac{V_1}{V_m} - 1 \right] = R_m \left[\frac{V_1}{I_m \times R_m} - 1 \right]$$

$$R_1 = 1K \left(\frac{5}{(50 \mu \times 1K)} - 1 \right) = 99 K\Omega$$

$$R_2 = 1K \left(\frac{10}{(50 \mu \times 1K)} - 1 \right) = 199 K\Omega$$

$$R_3 = 1K \left(\frac{50}{(50 \mu \times 1K)} - 1 \right) = 999 K\Omega$$

Note: In case of ammeter full scale current is minimum then ammeter is more sensitive

Ex: An ammeter with 1 mA of full scale current is more sensitive than an ammeter with 2 mA full scale current.

- The internal resistance of voltmeter is very high and ideally it is infinite
- In all the electrical instruments the responsible quantity for measurement is current
- In all electronic instruments the responsible quantity is voltage

Note: If ammeter is connected in parallel it will damage. If voltmeter is connected in series it won't damage because of its high internal resistance

- If voltmeter connected in series with load, it shows almost supply volts (full deflection) when load resistance is less and ammeter reads currents through voltmeter as shown fig. 1.

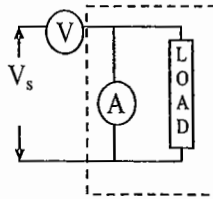


Fig (1)

Here ammeter and load in parallel $m\Omega \parallel M\Omega = m\Omega$ only. So total parallel combination is order of ' $m\Omega$ '.

So voltmeter is almost full scale

$$V \approx V_s$$

$$A \approx 0$$

For the below connection

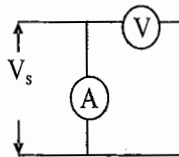


Fig (2)

Ammeter will damage and It becomes short circuit. So no voltage will read across (V)

So A = damaged

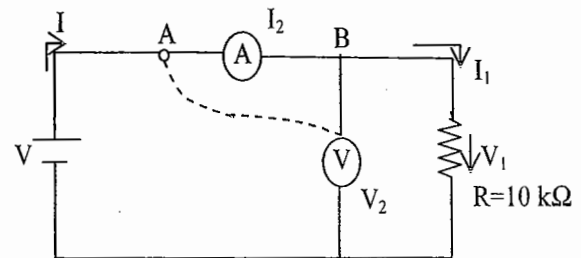
$$V \approx \text{zero}$$

Note: Basically there is no constructional difference between dc ammeter and dc voltmeter. If internal resistance of any ammeter is known ammeter can be converted as voltmeter (or) voltmeter can be converted as ammeter.

Note: PMMC measures average value and MI instrument will measure effective value (or) rms value.

Example 5:

Which of the following statements to determine the value of resistance R in the circuit shown is true?



- The voltage V_2 indicated by the voltmeter is higher than the voltage V_1 across the consumer.
- The Indicated current I_1 is smaller than I
- The voltage V_2 indicated by the voltmeter is less than the voltage V_1 across the consumer
- For a more accurate result, the voltmeter should be connected in front of the ammeter (Point A)

Ans: (d)

Tips to Solve a Voltmeter with different Ratings

- If voltmeter reading is given in 'V and S_v

$$R_s + R_m = S_v \times V$$

$$\frac{R_s + R_m}{V} = S_v$$

- If ammeter ratings are given in 'R' I_{fs}

$$R_s + R_m = R$$

- V, I_{fs} are given then

$$R_s + R_m = \frac{V}{I_{fs}}$$

$R_s \rightarrow R_{se}$ series multiplier resistance

R_m meter resistance



I_{fs} full scale current

From the above relations find individual meters resistor values 'R' then apply voltage division rule for finding the voltage measured by each meter

Loading effect: If a "Low sensitivity" voltmeter is used to measure the voltage across resistance. Then some of current will be passing through the voltmeter. So that meter reads wrong value this is called loading effect.

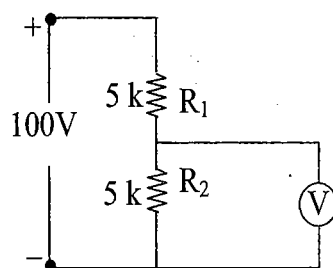
To reduce the loading effect high sensitivity voltmeters should be used

If the output of half wave rectifier is passing through PMMC then it will measure $I_{avg} = \frac{I_m}{\pi}$

If the output full wave rectifier is passing through PMMC. Then it will measure $I_{avg} = \frac{2I_m}{\pi}$

Example 6:

In the circuit shown in figure, voltage across R_2 is measured by two different voltmeters P and Q having sensitivity of $1K\Omega/V$ and $20K\Omega/V$. Both the meter are used on 50V scale, which of the following statement is true



- Voltmeter P will read more accurate than voltmeter Q
- Both the voltmeters read same voltage
- Voltmeter Q will read more accurate than voltmeter P.
- None of above

Ans: (c)

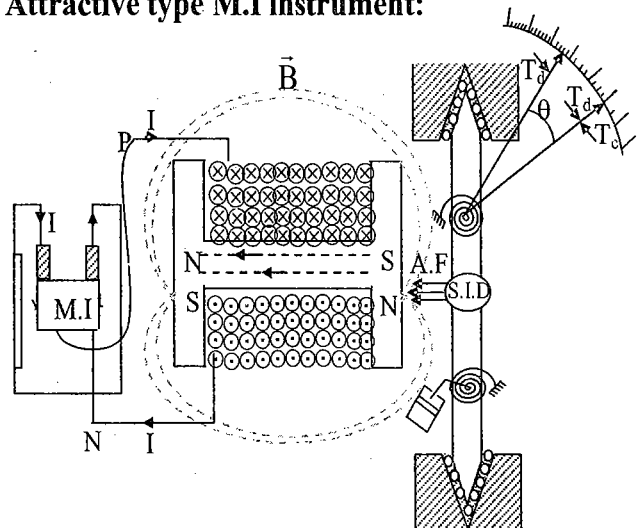
Sol: Voltmeter with high sensitivity will give more accurate results.

Moving Iron Instruments

These are two types:

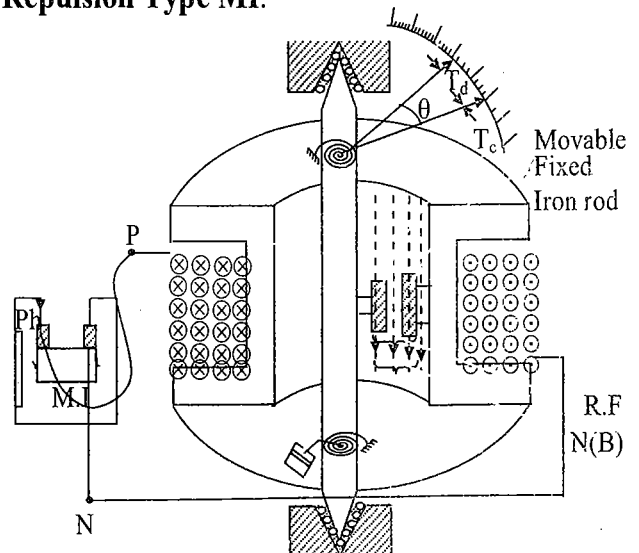
- Attractive type MI instrument
- Repulsion type MI instrument

Attractive type M.I instrument:



- When a current is passed through the coil, coil will setup a magnetic field, which attracts the iron disc, then pointer gives the deflections. This is attracted type M.I.

Repulsion Type MI:





- When the moment switch is closed, the current will enter and magnetic field is formed in hollow cylinder handle. Due to the presence of magnetic field, the repulsion force, developed due to the force spindle will rotate.

Note: The deflecting torque is derived in transient period only i.e. in 'dt' sec.

$$L = \frac{N\phi}{I}$$

$$LI = N\phi$$

$$\frac{d}{dt}(LI) = \frac{d}{dt}(N\phi)$$

$$I \cdot \frac{dL}{dt} + L \frac{dI}{dt} = N \frac{d\phi}{dt} = V$$

Electrical energy supplied to coil in dt sec.

$$\begin{aligned} VI dt &= \left(I \cdot \frac{dL}{dt} + L \frac{dI}{dt} \right) I dt \\ &= I^2 dL + LI dI \text{ -----(1)} \end{aligned}$$

$$\text{Initial energy stored} = \frac{1}{2} L \cdot I^2$$

Change in stored energy in 'dt' sec

$$\begin{aligned} &= \frac{1}{2} [L \cdot 2I dI + I^2 dL] \\ &= LI dL + \frac{1}{2} I^2 dL \text{ -----(2)} \end{aligned}$$

Mechanical work done in 'dt' sec

$$= T_d \cdot d\theta \text{ ---(3)}$$

Energy conservation law

$$(1) = (2) + (3)$$

$$I^2 dL + LI dL = LI dL + \frac{1}{2} I^2 dL + T_d \cdot d\theta$$

$$\frac{1}{2} I^2 dL = T_d \cdot d\theta$$

$$T_d = \frac{1}{2} I^2 \frac{dL}{d\theta}$$

$$\text{If } \frac{dL}{d\theta} \text{ is constant then } T_d \propto I^2$$

Spring control is used to produce controlling torque i.e. $T_c = K_c \theta$

Air friction damping is used to produce Damping torque.

Note: Eddy current damping is not possible because of presence of Iron disc and poor magnetic field.

At final steady state position $T_c = T_d$

$$\theta = \frac{1}{2} \cdot \frac{I^2}{K_c} \cdot \frac{dL}{d\theta}$$

$$\text{If } \frac{dL}{d\theta} \text{ is constant}$$

$$\theta \propto I_{rms}^2$$

M.I instrument has non-linear sequence (or) sequence law scale.

Cramped (very close) at beginning & no number of division at the full scale.

M.I instruments works for A.C and D.C.

Note: For AC even though current direction reversed magnetic field direction reversed but always attractive force is present between coil magnetic field & iron disc. i.e. unidirectional torque.

Note: MI will give wrong values for both AC & DC. For DC MI instrument will give accurate values, hence MI is not used for DC.

Note: There is no alternate instrument for AC.

To measure correctly, Hence MI is used only measure AC & scale is calibrated in terms of R.M.S value.

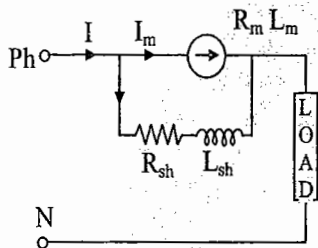


MI instrument can be used as Ammeter up to 50A measurements only & as a voltmeter up to 750V only. Because of the magnetic saturation of coil. If the O/P of H.W.R is passing through M.I. It

measures $\frac{I_m}{2}$.

If the O/P of F.W.R is passing through M.I. It measures $\frac{I_m}{\sqrt{2}}$.

To increase range of ammeter, extra number of turns are connected in parallel with the meter



If, $x_L = 2\pi fL$, $Z_m = R_m + jx_m$

$$Z = \sqrt{R^2 + x^2}$$

$$I_{sh} Z_{sh} = I_m Z_m$$

$$\frac{I_{sh}}{I_m} = \frac{Z_m}{Z_{sh}} = \frac{\sqrt{R_m^2 + (2\pi fL_m)^2}}{\sqrt{R_{sh}^2 + (2\pi fL_{sh})^2}}$$

$$\frac{\sqrt{R_m^2 + (\omega L_m)^2}}{\sqrt{R_{sh}^2 + (\omega L_{sh})^2}} = \frac{R_m \sqrt{1 + \left(\frac{\omega L_m}{R_m}\right)^2}}{R_{sh} \sqrt{1 + \left(\frac{\omega L_{sh}}{R_{sh}}\right)^2}}$$

$$\text{If } \frac{L_{sh}}{R_{sh}} = \frac{L_m}{R_m}$$

$$\frac{I_{sh}}{I_m} = \frac{R_m}{R_{sh}}$$

Frequency Independent. Error due to frequency is eliminated.

If shunt time constant = meter time constant. Then division of current in meter and shunt is independent of frequency and depends only on resistance. It is very difficult practically

Example 7:

A moving iron voltmeter calibrated for 50 Hz supply and having an inductance of the coil as 0.8 H draws 0.1 A when connected to 500 V a.c. What will be its reading when connected to 300 V dc _____ (in V)

Ans: 300.4

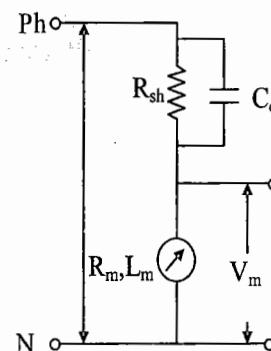
$$\text{Sol: } Z = \frac{V_{ac}}{I_{ac}} = \frac{500}{0.1} = 5000 \Omega$$

$$R = \sqrt{(5000)^2 - (2\pi \times 50 \times 0.8)^2} = \sqrt{(5000)^2 - (251.327)^2} = 4993.7 \Omega$$

$$I_{dc} = \frac{V_{dc}}{R} = \frac{300}{4993.7} = 0.06008 \text{ A}$$

$$\therefore \text{Reading of voltmeter when connected for 300V dc} = \frac{500}{0.1} \times 0.06008 = 300.4 \text{ V}$$

Error in voltmeter:



To reduce the frequency error a compensating capacitor is connected across multiple resistance (R_{se})



$$\text{Compensating capacitor } (c_c) = \frac{0.41 L m}{R_{se}^2}$$

Note: There is a physical constructional difference between AC ammeter and AC voltmeter.

AC ammeter has thick cross-section and few number of turns (low resistance)

$$\downarrow R_a = \frac{\rho \cdot \ell (\downarrow) \rightarrow \text{few turns}}{a (\uparrow) \text{ more area}}$$

Note: And AC voltmeter has more number of turns with thin cross section (high resistance)

$$\uparrow R_v = \frac{\rho \cdot \ell \rightarrow \text{more turns}}{a \rightarrow \text{Less area}}$$

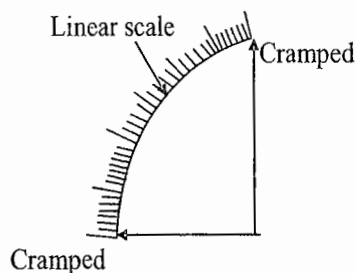
$$\text{If } f \uparrow = X_L \uparrow$$

To reduce the Increased reactance of L we have to connect a capacitor for decrease the increased reactance

$$\text{If } f \uparrow = (X_L \uparrow - X_C \downarrow)$$

Note: We cant connect capacitor in series, it forms series RLC as 'f' increases X_L & X_C gets compensated X cancelled. The CKT (meter) then becomes only connected with R. & It will damage for high currents. So we have to connect capacitor in parallel to R_{sh}

Condition for linear scale:



$$T_d = T_c$$

$$\frac{1}{2} I^2 \cdot \frac{dL}{d\theta} = k\theta$$

$$\text{Linear scale } I \propto \theta \quad I = K_1 \theta$$

$$\frac{1}{2} (K_1 \theta)^2 \frac{dL}{d\theta} = k\theta$$

$$\theta^2 \cdot \frac{dL}{d\theta} = 2 \frac{k}{K_1} \theta$$

$$\theta \frac{dL}{d\theta} = 2 \frac{k}{K_1} = \text{constant}$$

→ Condition for linear scale

Note: To measure very high values of AC currents and A.C voltages, Instrument Transformers are used i.e C.T (Current Transformers) and P.T (Potential Transformer) Because of cost point of view of potential transformers, we are using capacitive voltage transformers are used in electrical substations

To convert M.I ammeter into voltmeter is same as in converting PMMC instruments.

Example 8:

A 0 – 200mA moving iron ammeter is converted to a 0-600V, 50Hz voltmeter by adding a series resistance with the coil. The coil has negligible resistance and an inductance $L = (0.1 + 0.2\theta) / 4\pi$ henry, where θ is the deflection in radian. The total angular span of the meter is 1000. Find the series resistance?

- (a) 11.22 Ω (b) 0.6 k Ω
(c) 1.2k Ω (d) 3 k Ω

Ans: (d)

Sol:

The impedance required to convert the meter to 600V voltmeter.